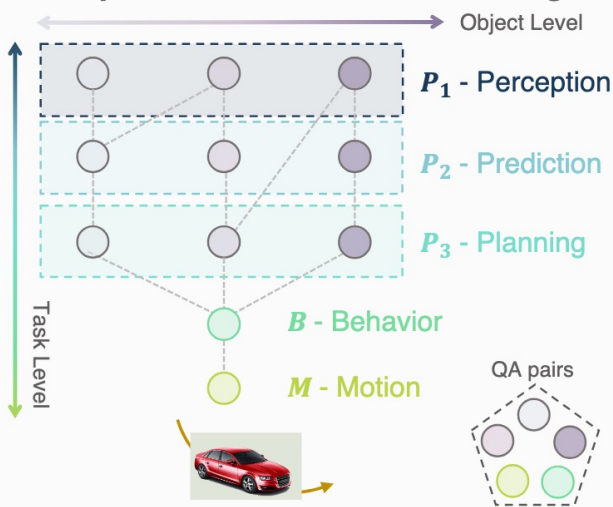


# DriveLM: Driving with Graph Visual Question Answering

Chonghao Sima\*, Katrin Renz\*, Kashyap Chitta, Li Chen,  
Hanxue Zhang, Chengen Xie, Jens Beißwenger,  
Ping Luo, Andreas Geiger, Hongyang Li

## Graph Visual Question Answering



Q – Question; A – Answer;

## DriveLM-Data

$C$ : Context

### $P_1$ - Perception

Q: What are the **objects** worth noting in the current scenario?

A: There is a **van**, a **sedan** and a **pedestrian** in front of the ego car.

### $P_2$ - Prediction

Q: Where might the **van**, the **sedan** and the **pedestrian** move in the future?  $C: P_1$

A: Those **objects** may move to the right lane, cross the ped-crossing, etc..

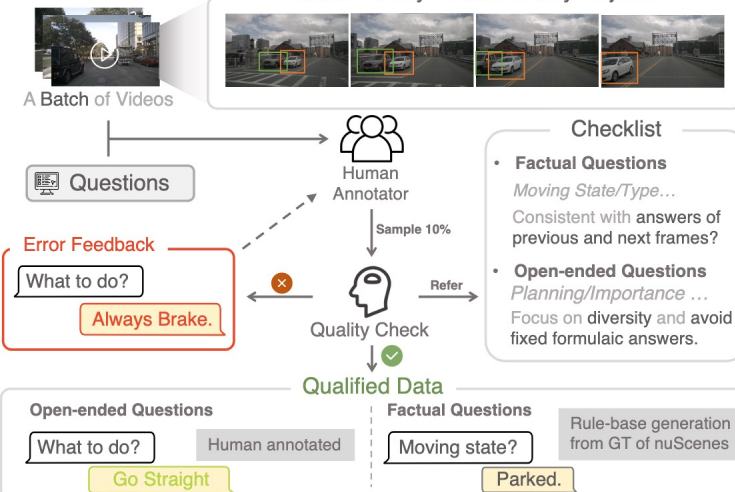
### $P_3$ - Planning

Q: What are the **safe actions** of the ego car considering those **objects**?  $C: P_2$

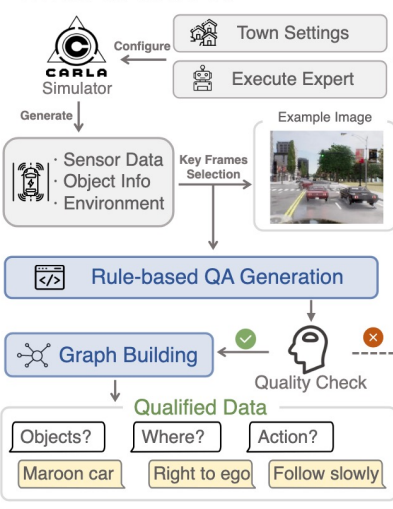
A: The ego car should **wait** for the **pedestrian** and the **sedan**, then **turn right**.

## DriveLM-nuScenes

Selected Key Frames & Key Objects



## DriveLM-CARLA



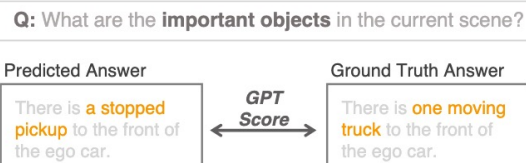
## Dataset Comparision

| Dataset          | Source Dataset | # Frames | Avg. QA per Frame |
|------------------|----------------|----------|-------------------|
| nuScenes-QA [45] | nuScenes       | 34,149   | 13.5              |
| nuPrompt [67]    | nuScenes       | 34,149   | 1.0               |
| HAD [27]         | HDD            | 25,549   | 1.8               |
| BDD-X [28]       | BDD            | 26,228   | 1                 |
| LingoQA [38]     | LingoQA        | 28,000   | 15.3              |
| DRAMA [35]       | DRAMA          | 17,785   | 5.8               |
| Rank2Tell [48]   | Rank2Tell      | 5,800    | -                 |
| DriveLM-nuScenes | nuScenes       | 4,871    | 91.4              |
| DriveLM-CARLA†   | CARLA          | 64,285   | 24.4              |
| DriveLM-CARLA‡   | CARLA          | 5,721    | 24.8              |

For more research update:  
[opendrivelab.com](https://opendrivelab.com) [cvlibs.net](https://cvlibs.net)

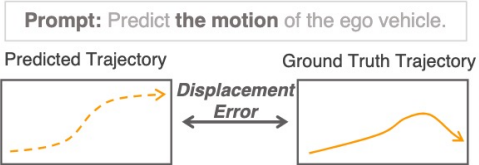


## Graph Visual QA



## DriveLM-Metrics

### Trajectory Prediction



## Generalization

